
Raw HEIC/JPG Images
(4284×5712)
Format Conversion
HEIC → Standard
Resize
(256×300)
Center Crop
(128×128)
Grayscale Conversion
RGB → Grayscale
Data Augmentation
Flip, Affine, Photometric
Normalization
to [-1, 1]
Final Tensor
(1, 128, 128)

Figure 1: Data Preprocessing Pipeline: Complete workflow from raw images through format conversion, geometric transformations, augmentation, and normalization to produce training-ready tensors.

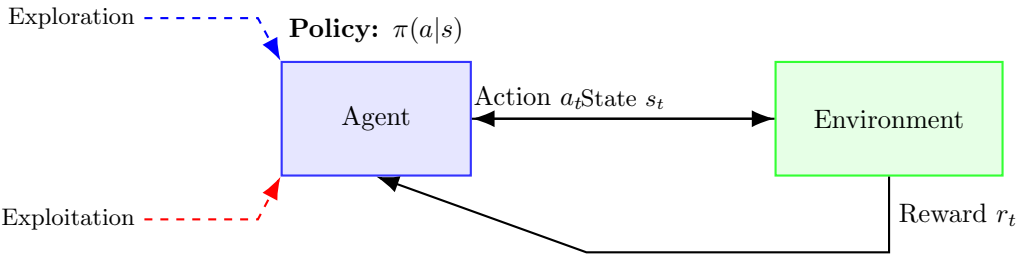

Exploration
Policy: $\pi(a|s)$
Agent
Action a_tState s_t
Environment
Exploitation
Reward r_t

Figure 2: Reinforcement Learning Agent Interacting with Environment: Balance of Exploration and Exploitation

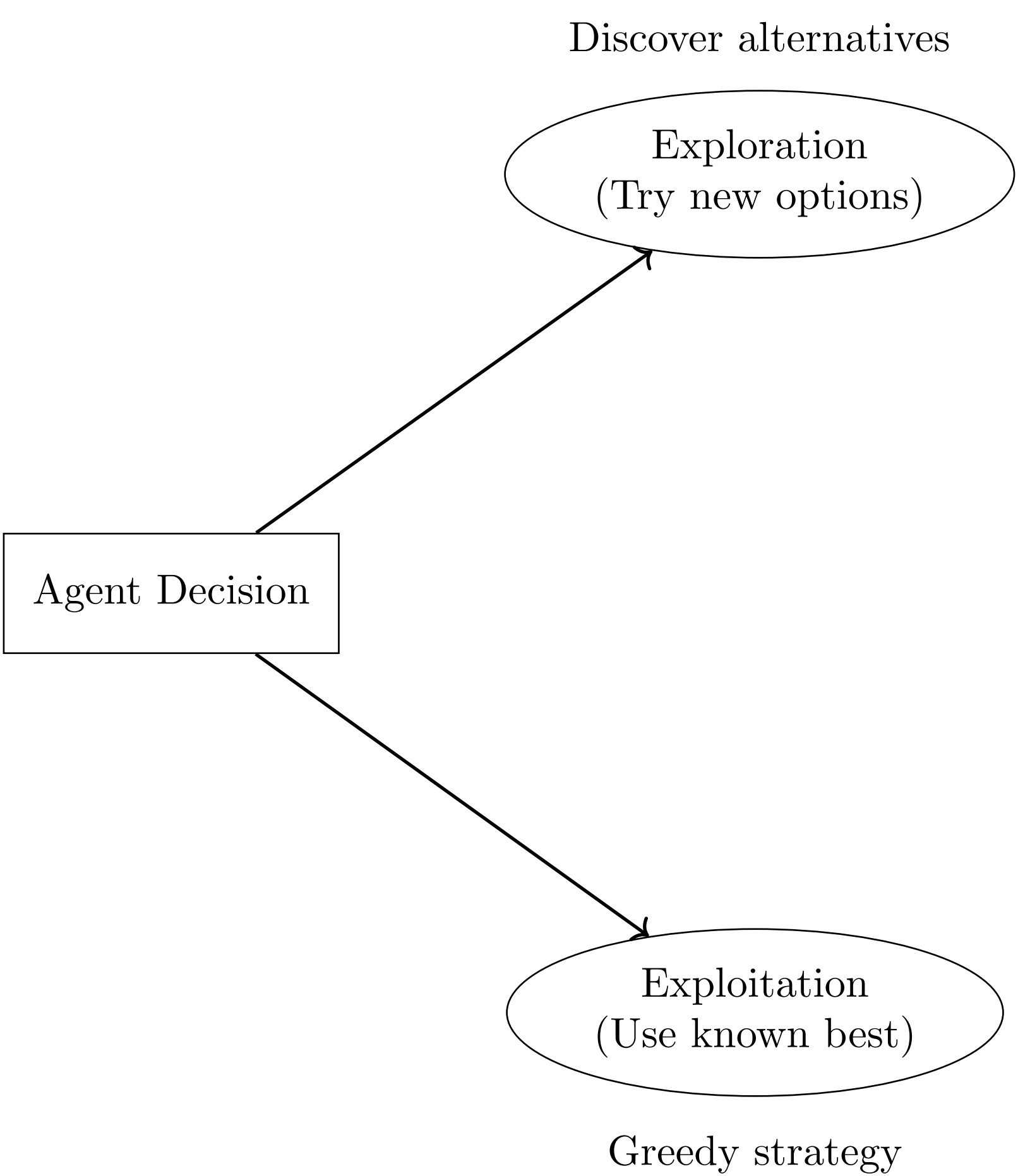

Discover alternatives
Exploration
(Try new options)
Agent Decision
Exploitation
(Use known best)
Greedy strategy

Figure 3: Exploration vs. Exploitation decision in agent learning

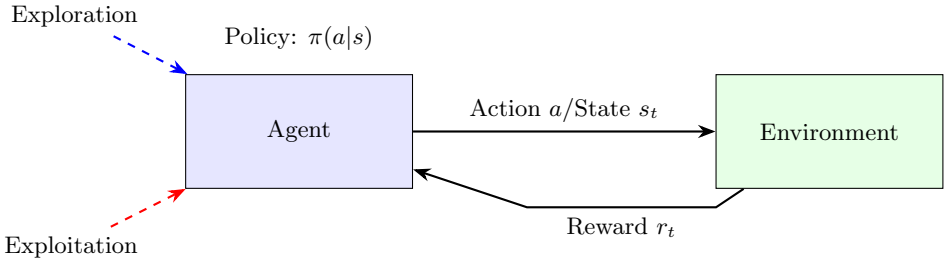

Exploration
Policy: $\pi(a|s)$
Agent
Action a/State s_t
Environment
Reward r_t
Exploitation

Figure 4: Reinforcement Learning Agent Interacting with Environment: Balance of Exploration and Exploitation

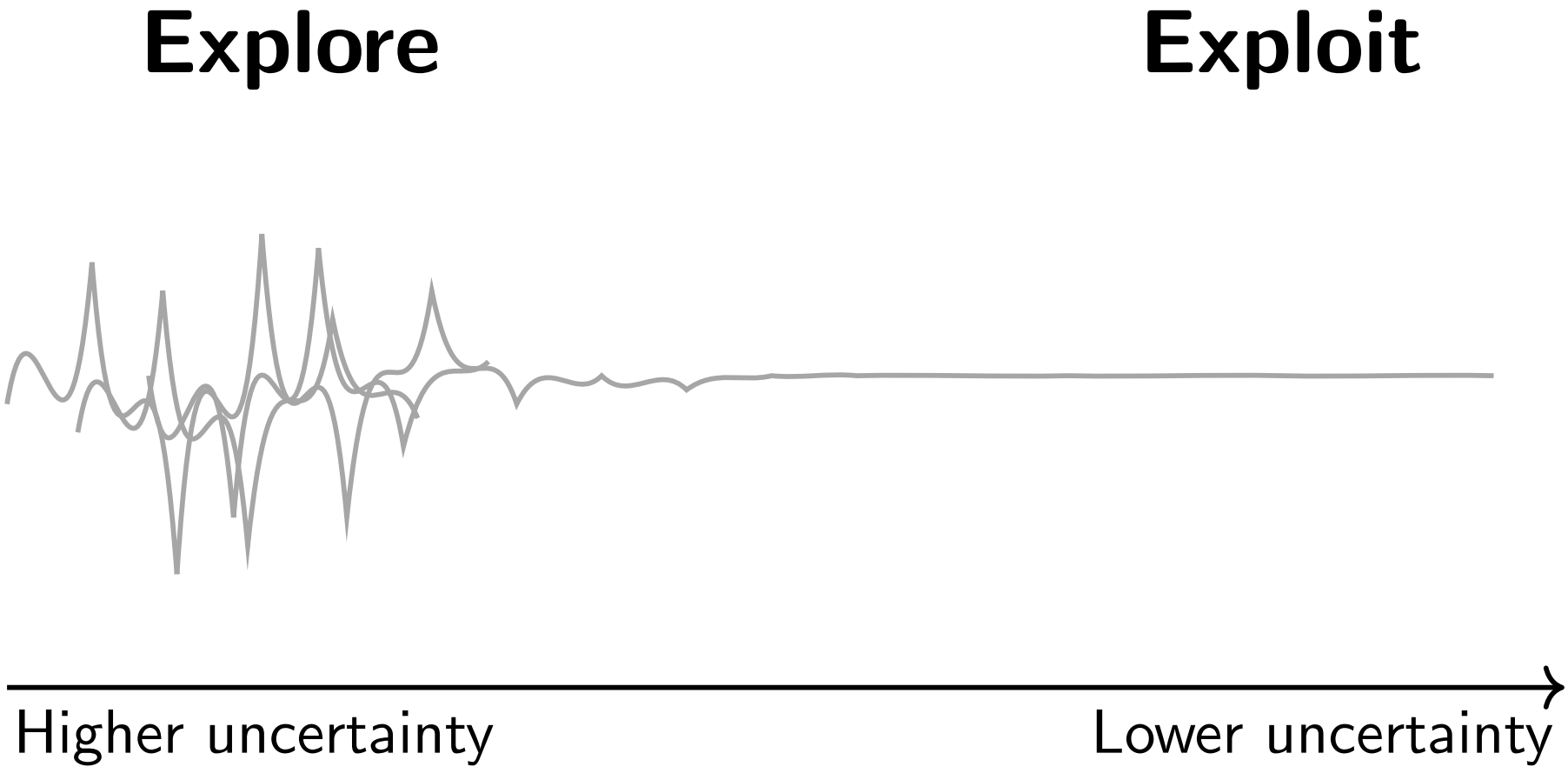

Explore
Exploit
Higher uncertainty
Lower uncertainty

Figure 5: Explore–Exploit Trade-off: Random exploration under high uncertainty transitioning into stable exploitation.

0.1 State, Action, Reward Design

State Space: The state s_t represents the current condition of the GA population. It is described using a set of summary features:

- Best fitness score in generation t
- Average fitness of the population
- Diversity metric (e.g., Hamming distance between chromos

RL-GA Meta-Optimization Framework

